CS 486/686
Dissecting a Vision-Language Model

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Lecture 23

How an image becomes language-model context

Search › Uncertainty › Decisions › Learning

Learning goals

- Trace **image + text** inference end to end.
- Explain **visual tokens** and **vision encoders**.
- Explain how visual features enter a **language decoder**.
- Recognize VLM tasks and failure modes.

From text-only to multimodal

L21

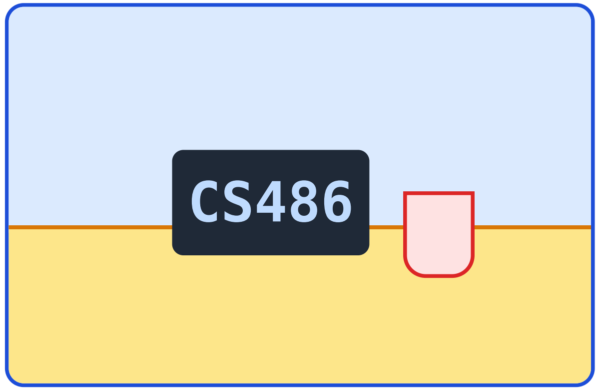
text → tokens → LM → answer

L23

image + text → visual + text tokens → answer

A VLM is a language model with extra visual context.

Running example




```
What is shown in
this image?
```

The model must combine visual evidence with language generation.

What makes this hard?

pixels

huge arrays

space

details have positions

language

answer must be text

The VLM pipeline

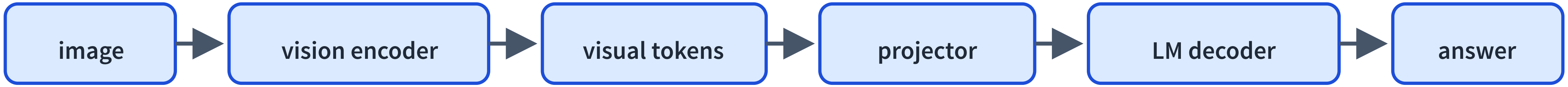


Image information becomes context the language model can use.

What gets loaded?

processor

image preprocessing

tokenizer

text tokens

vision encoder

visual features

decoder

generates text

Demo path and fallback

optional live path

Qwen3-VL +
Transformers.js/WebGPU

reliable path

precomputed traces in slides

Optional browser code sketch

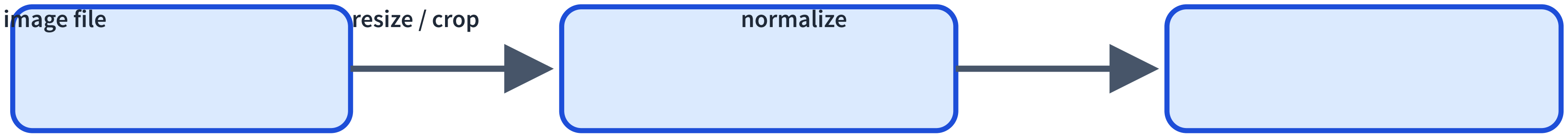
```
const processor = await AutoProcessor.from_pretrained(model_id);
const model = await Qwen3VLForConditionalGeneration.from_pretrained(model_id, {
  device: "webgpu",
  dtype: { vision_encoder: "fp16", decoder_model_merged: "q4f16" },
});
```

Conceptual code only: live VLM inference depends on browser and GPU support.

The same idea in Python

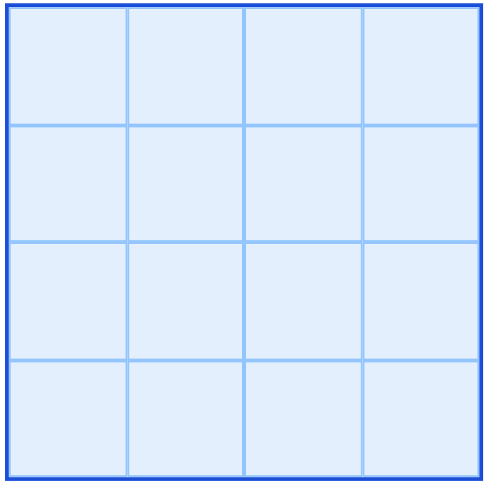
```
processor = AutoProcessor.from_pretrained("Qwen/Qwen3-VL-2B-Instruct")
model = Qwen3VLForConditionalGeneration.from_pretrained(
    "Qwen/Qwen3-VL-2B-Instruct",
    dtype="auto",
    device_map="auto",
)
```

Images must become tensors



Raw image files are not what the network consumes.

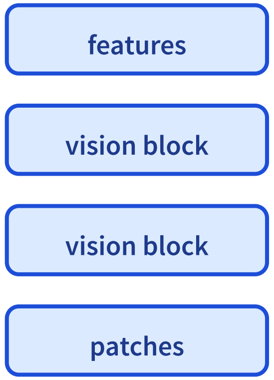
Patchify the image



A vision transformer treats patches like a sequence.

Patch vectors are visual tokens.

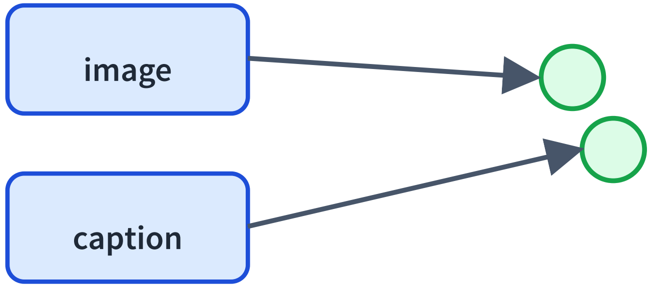
The vision encoder extracts visual features



The image is processed before it reaches the language model.

Qwen-style VLMs use a dedicated vision encoder and a merger/projector.

Why CLIP matters



CLIP-style training aligns image and text embeddings.

It is the bridge from visual features to language semantics.

CLIP is not a full VLM

CLIP

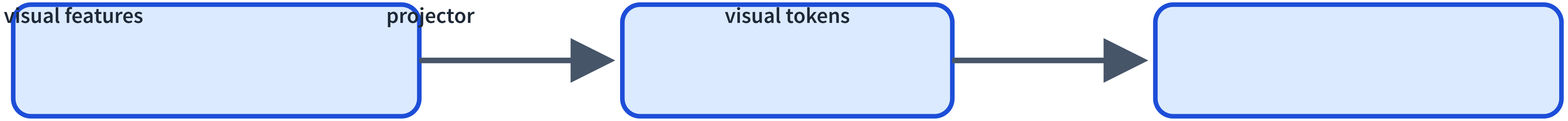
compare image/text embeddings

VLM

generate text conditioned on image tokens

The projector translates vision to LM space

visual features → MLP merger/projector → LM-sized visual tokens



Multimodal context sequence

`img1` `img2` `img3` `What` `is` `shown?` `A` `laptop`

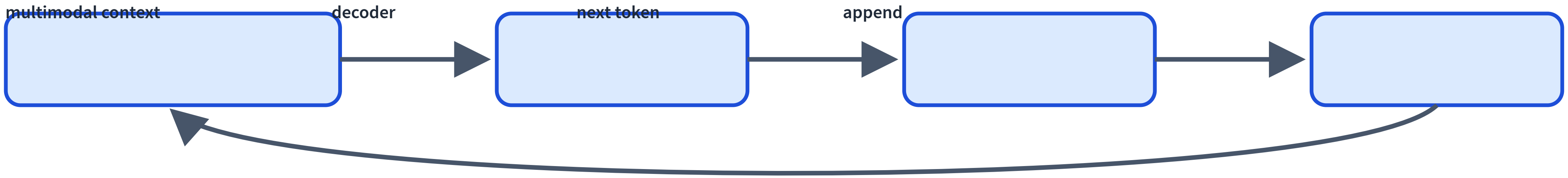
The decoder can attend to image tokens and text tokens in the same context.

Chat template with an image placeholder

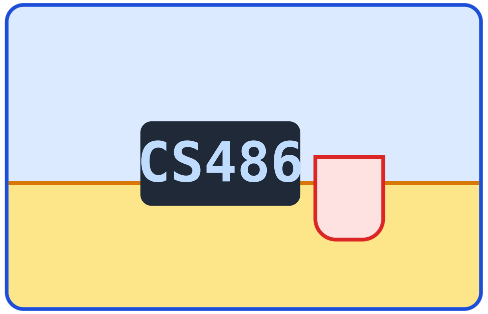
user	[image]
user	Describe this image.
assistant	

The prompt is structured multimodal context, not just pixels.

Generation is still autoregressive



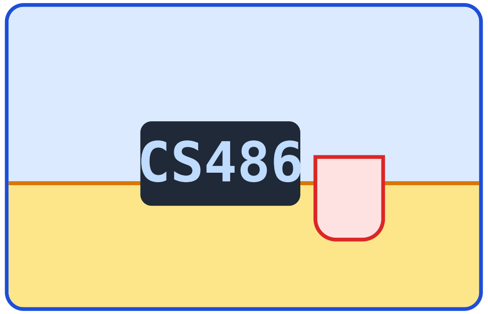
Trace: captioning




```
Describe this image.
```

A laptop and cup are sitting on a desk.

Trace: visual question answering




```
What object is next
to the laptop?
```

A cup is next to the laptop.

Trace: document/OCR understanding

Course Schedule

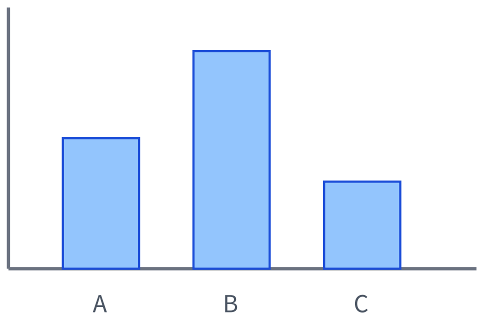
Assignment 2 due Thu Jul 9

Chat 9 out Thu Jul 9

```
When is Assignment 2 due?
```

Assignment 2 is due Thu Jul 9.

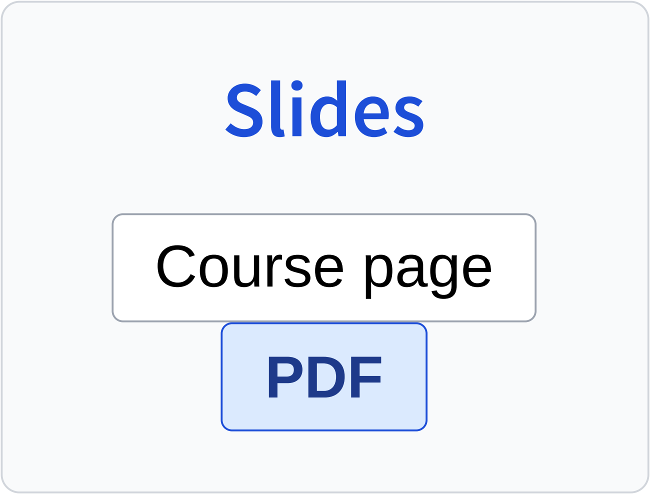
Trace: chart understanding




```
Which bar is
largest?
```

Bar B is largest.

Trace: UI/screenshot understanding

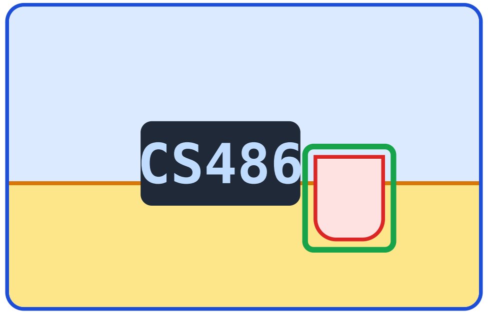



```
Which button
downloads the
slides?
```

Click the PDF button.

Some VLMs can ground answers spatially

Instead of only text, a model may output a box or point.



Grounding is useful for documents, UI agents, and object localization.

What gets trained?

vision encoder	**projector**	**decoder**
visual features	align dimensions	language generation

VLM ability depends on multimodal data

captions

image-text pairs

documents

OCR/forms/tables

screens

UI/action traces

Failure mode: hallucinated visual details

The model may describe objects that are not present.

Fluent visual descriptions are not guaranteed faithful.

Failure mode: OCR errors

The image says **Jul 9**; the model reads **Jul 8**.

Reading text in images is powerful but brittle.

Failure mode: spatial reasoning

left/right

easy to flip

counting

small objects

relations

above/below

Failure mode: prompt sensitivity

describe

Describe what you see.

guess

Guess what is happening.

Visual answers can change with linguistic framing.

VLMs enable visual agents

observe screenshot → reason → click/type → observe again

Same tool/agent safety concerns from L22 still apply.

What did we add beyond text LMs?

L21

text tokens only

L23

image tokens + text tokens

Both generate text autoregressively.

Next question

Now models can understand images.

How do models generate images or future frames?

L24: Dissecting Diffusion and World Models.

Recap

✓ Images become tensors, patches, and visual tokens.

✓ A vision encoder extracts visual features.

✓ A projector maps visual features into LM space.

✓ The decoder attends to image and text context.

✓ VLMs can caption, answer, read, and reason visually.

✓ Failure modes include hallucination, OCR, spatial mistakes, and prompt sensitivity.